

Full Test - Sets, Relations and Functions

Time Allowed : 1 hour _____ Maximum Marks : 200

Instructions

1. The test consists of forty Questions
2. Each question carries +5 Marks for a correct response and -2 marks for an incorrect response.

Q1 Let $R = (3, 3)(6, 6)((9, 9)(12, 12), (6, 12)(3, 9)(3, 12), (3, 6)$ be a relation on the set $A = 3, 6, 9, 12$. The relation is

- (a) reflexive and transitive
- (b) reflexive only
- (c) an equivalence relation
- (d) reflexive and symmetric only

Q2: Let $f : (-1, 1) \rightarrow B$ be a function defined by $f(x) = \tan^{-1} \frac{2x}{1-x^2}$, then f is both one-one and onto when B is the interval

- (a) $(0, \frac{\pi}{2})$ (b) $[0, \frac{\pi}{2}]$ (c) $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (d) $(-\frac{\pi}{2}, \frac{\pi}{2})$

Q3: If a real valued function $f(x)$ satisfies the functional equation

$f(x-y) = f(x)f(y) - f(a-x)f(a+y)$, where 'a' is a given constant and $f(0) = 1$, then $f(2a-x)$ is equal to

- (a) $-f(x)$
- (b) $f(x)$
- (c) $f(a) + f(a-x)$
- (d) $f(-x)$

Q4: Let $R = (1, 3), (4, 2), (2, 4), (2, 3), (3, 1)$ be a relation on the set $A = 1, 2, 3, 4$. The relation R is

- (a) a function
- (b) transitive
- (c) not symmetric
- (d) reflexive

Q5: The range of the function $f(x) = {}^{7-x}P_{x-3}$ is

- (a) 1, 2, 3
- (b) 1, 2, 3, 4, 5, 6
- (c) 1, 2, 3, 4
- (d) {1, 2, 3, 4, 5}

Q6: If $f : R \rightarrow S$, defined by $f(x) = \sin x - \sqrt{3} \cos x + 1$ is onto, then the interval of S is

- (a) $[0, 3]$
- (b) $[-1, 1]$
- (c) $[0, 1]$
- (d) $[-1, 3]$

Q7: The graph of the function $f(x)$ is symmetrical about the line $x = 2$, then

- (a) $f(x+2) = f(x-2)$
- (b) $f(2+x) = f(2-x)$
- (c) $f(x) = f(-x)$
- (d) $f(x) = -f(-x)$

Q8: The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is

- (a) $[2, 3]$
- (b) $[23]$
- (c) $[1, 2]$
- (d) $[1, 2)$

Q9: If $f : 1, 2, 3, \dots \rightarrow \{0, \pm 1, \pm 2, \dots\}$ is defined by

$$f(x) = \begin{cases} \frac{x}{2}, & \text{if } x \text{ is even} \\ -\frac{(x-1)}{2}, & \text{if } x \text{ is odd} \end{cases} \quad \text{then value of } f^{-1}(-100) \text{ is}$$

- (a) 100
- (b) 199
- (c) 200
- (d) 201

Q10: Domain of definition of the function $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is

- (a) $(1, 2)$
- (b) $(-1, 0) \cup (1, 2)$
- (c) $(1, 2) \cup (2, \infty)$
- (d) $(-1, 0) \cup (1, 2) \cup (2, \infty)$

Q11: The function $f(x) = \log(x + \sqrt{x^2 + 1})$ is a/an

- (a) even function
- (b) odd function
- (c) periodic function
- (d) none of these

Q12: The function $f : R \rightarrow R$ defined by $f(x) = \sin x$ is

- (a) into
- (b) onto
- (c) one-one
- (d) many-one

Q13: The range of the function $f(x) = \frac{2+x}{2-x}$, $x \neq 2$ is

- (a) R
- (b) $R - \{-1\}$
- (c) $R - \{1\}$
- (d) $R - \{2\}$

Q14: If $f(x) = \begin{cases} x & , x \in Q \\ 0 & , x \notin Q \end{cases}$ and $g(x) = \begin{cases} 0 & , x \in Q \\ x & , x \notin Q \end{cases}$ then $(f - g)$ is

- (a) one-one, onto
- (b) neither one-one nor onto
- (c) one-one but not onto
- (d) onto but not one-one

Q15: If $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$, then $g[f(x)]$ will be invertible for the domain

- (a) $[0, \pi]$
- (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
- (c) $\left[0, \frac{\pi}{2}\right]$
- (d) $\left[-\frac{\pi}{2}, 0\right]$

Q16: The range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$, $x \in (-\infty, \infty)$ is

- (a) $[1, \infty)$
- (b) $\left(1, \frac{11}{7}\right)$
- (c) $\left(1, \frac{7}{3}\right]$
- (d) $\left[1, \frac{7}{5}\right]$

Q17: $f : [0, \infty) \rightarrow [0, \infty)$, $f(x) = \frac{x}{1+x}$ is

- (a) one-one and onto
- (b) one-one but not onto
- (c) onto but not one-one
- (d) neither one-one nor onto

Q18: If $f(x) = (x+1)^2$ for $x \geq -1$ and $g(x)$ is the function whose graph is reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals

- (a) $-\sqrt{x} - 1$, $x \geq 0$
- (b) $\frac{1}{(x+1)^2}$, $x > -1$
- (c) $\sqrt{x+1}$, $x \geq -1$
- (d) $\sqrt{x-1}$, $x \geq 0$

Q19: If function $f : R \rightarrow R$ is defined as $f(x) = 2x + \sin x$ for $x \in R$, then f is

- (a) one-one and onto
- (b) one-one but not onto
- (c) onto but not one-one
- (d) neither one-one nor onto

Q20: If $g(x) = 1 + x - [x]$ and $f(x) = \tan^{-1} x$ then for all x , $f[g(x)] =$

- (a) x
- (b) 1
- (c) $f(x)$
- (d) $g(x)$

Q21: If $f : [1, \infty) \rightarrow [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals

- (a) $\frac{x + \sqrt{x^2 - 4}}{2}$
- (b) $\frac{x}{1 + x^2}$
- (c) $\frac{x - \sqrt{x^2 - 4}}{2}$
- (d) $1 + \sqrt{x^2 - 4}$

Q22: The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$ is

- (a) $R - \{-1, -2\}$
- (b) $(-2, \infty)$
- (c) $R - \{-1, -2, -3\}$
- (d) $(-3, \infty) - \{-1, -2\}$

Q23: If $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$, then the number of onto functions from E to F is

- (a) 14
- (b) 16
- (c) 12
- (d) 8

Q24: If $f(x) = \frac{ax}{x+1}$, $x \neq -1$, then for which value of a is $f[f(x)] = x$?

- (a) $\sqrt{2}$
- (b) $-\sqrt{2}$
- (c) 1
- (d) -1

Q25: Let $f : R \rightarrow R$ be any function. Define $g : R \rightarrow R$ by $g(x) = |f(x)|$ for all x . Then g is

- (a) onto if f is onto
- (b) one-one if f is one-one
- (c) continuous if f is continuous
- (d) differentiable if f is differentiable

Q26: The domain of definition of the function $y(x)$ as given by the equation $2^x + 2^y = 2$ is

- (a) $0 < x \leq 1$
- (b) $0 \leq x \leq 1$
- (c) $-\infty < x \leq 0$
- (d) $-\infty < x < 1$

Q27: If the function $f : [1, \infty) \rightarrow [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is

- (a) $\left(\frac{1}{2}\right)^{x(x-1)}$
- (b) $\frac{1}{2}(1 + \sqrt{1 + \log 2x})$
- (c) $\frac{1}{2}(1 + \sqrt{1 - 4\log 2x})$
- (d) not defined

Q28: In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is

- (a) at least 30
- (b) at most 20
- (c) exactly 25
- (d) none of these

Q29: If $f(x) = \frac{x^2 - 1}{x^2 + 1}$, for every real number x , then the minimum value of f

- (a) does not exist as f is unbounded
- (b) is equal to 1
- (c) is not attained even though f is bounded
- (d) is equal to -1

Q30: If $f(x) = 3x - 5$, then $f^{-1}(x)$

- (a) is given by $\frac{1}{3x - 5}$
- (b) is given by $\frac{x + 5}{3}$
- (c) does not exist because f is not one-one
- (d) does not exist because f is not onto

Q31: If $g[f(x)] = |\sin x|$ and $f[g(x)] = (\sin \sqrt{x})^2$, then

- (a) $f(x) = \sin^2 x, g(x) = \sqrt{x}$
- (b) $f(x) = \sin x, g(x) = |x|$
- (c) $f(x) = x^2, g(x) = \sin \sqrt{x}$
- (d) f and g cannot be determined

Q32: If $f(x) = (x+1)^2 - 1, (x \geq -1)$, then the set $S = \{x : f(x) = f^{-1}(x)\}$ is

- (a) $\left\{0, -1, \frac{-3+i\sqrt{3}}{2}, \frac{-3-i\sqrt{3}}{2}\right\}$
- (b) $\{0, -1, -1\}$
- (c) $\{0, -1\}$
- (d) empty

Q33: The number $\log_2 7$ is

- (a) an integer
- (b) a rational number
- (c) an irrational number
- (d) a prime number

Q34: If S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains

- (a) $\left(-\infty, -\frac{3}{2}\right)$
- (b) $\left(-\frac{3}{2}, -\frac{1}{4}\right)$
- (c) $\left(-\frac{1}{4}, -\frac{1}{2}\right)$
- (d) $\left(\frac{1}{2}, 3\right)$
- (e) none of these

Q35: If $y = f(x) = \frac{x+2}{x-1}$, then

- (a) $x = f(y)$
- (b) $f(1) = 3$
- (c) y increases with x for $x < 1$
- (d) f is a rational function of x

Q36: Let $f(x) = |x-1|$. Then

- (a) $f(x^2) = [f(x)]^2$
- (b) $f(x+y) = f(x) + f(y)$
- (c) $f(|x|) = |f(x)|$
- (d) None of these

Q37: The domain of definition of the function $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ is

- (a) $(-3, -2)$ excluding -2.5
- (b) $[0, 1]$ excluding 0.5
- (c) $[-2, 1]$ excluding 0
- (d) None of these

Q38: Which of the following functions is periodic?

- (a) $f(x) = x - [x]$ where $[x]$ denotes the largest integer less than or equal to the real number x
- (b) $f(x) = \sin \frac{1}{x}$ for $x \neq 0$, $f(0) = 0$
- (c) $f(x) = x \cos x$
- (d) None of these

Q39: If X and Y are two sets, then $X \cap (X \cup Y)^C$ equals

- (a) X
- (b) Y
- (c) ϕ
- (d) none of these

Q40: Let R be the set of real numbers. If $f : R \rightarrow R$ is a function defined by $f(x) = x^2$, then f is

- (a) injective but not surjective
- (b) surjective but not injective
- (c) bijective
- (d) none of these

ALPHA CLASSES -OMR SHEET

USE ONLY BLACK OR BLUE BALL POINT PEN
DO NOT USE PENCIL/INK PEN/GEL PEN

Name of the Candidate	ROLL NUMBER	QUESTION BOOKLET NUMBER
	00000	00000
	11111	11111
	22222	22222
	33333	33333
	44444	44444
	55555	55555
	66666	66666
	77777	77777
	88888	88888
	99999	99999
Signature of the Candidate		
Date of Examination		
Signature of Invigilator		

1	A	B	C	D
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3	A	B	C	D
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37	A	B	C	D
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